Quantifying the Consumer's Dependence on Different Information Sources on Acceptance of Autonomous Vehicles

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ABSTRACT
Recent academic research and industrial commitments highlight the potential of connected and autonomous vehicles (CAVs) in transforming the way we travel. However, there are some anticipated barriers associated with the widespread consumer adoption of CAVs, including but not limited to low user acceptance, cybersecurity, safety, and legislation. The existing literature is non-existent in capturing the impact of information received from multiple sources (peers, car dealers, media advertisements, and personal research) on the consumers' acceptance of CAVs while considering these barriers associated with CAV. In this direction, we quantify the impact of the multiple information sources on the individuals concerned and indifferent about the anticipated barriers of CAVs to boost their acceptance while utilizing a two-step econometric framework based on an online survey of 4,448 Tennesseans. The two-step modeling framework utilizes latent class analysis followed by a multivariate ordered logit model. Results indicated that the elderly, three or more person households, and residents interested in owning a CAV are more likely to be concerned about CAV barriers. In contrast, males, physically challenged, and residents willing to pay higher for CAV are more likely to be indifferent toward CAV barriers. Among concerned individuals, the elderly, females, and residents willing to pay more than $30k for a conventional car are more likely to rely on car dealers when purchasing a CAV. The results of this study are expected to assist automakers, technology companies, policymakers, and third-party agencies to advertise, promote and introduce CAVs effectively through appropriate information channel(s) to boost their consumer acceptance.

Keywords: Latent class analysis; Media influence; Peer to peer interaction; CAV acceptance; Multivariate ordered logit.

1. Introduction

Consumer purchase behavior is affected by several factors (Akalamkam and Mitra, 2018). It involves five different stages, i.e., problem recognition, pre-purchase information search, evaluation of alternatives, purchase, and post-purchase evaluation (Schiffman et al., 2016), the most important being information search from both decision-making marketing perspectives (Akalamkam and Mitra, 2018). For instance, literature on conventional automobile purchases vindicates the influence of pre-purchase information from multiple sources on consumers' buying behavior. Before purchasing a vehicle, such consumers typically search for vehicle-related information through various search channels (van Rijnsoever et al., 2009), as product involvement is related positively to the use of all available information sources (van Rijnsoever et al., 2012). For instance, about 95% of automobile buyers use the internet as an information source in the US. More than 95% of vehicle sales occur at the dealership (Google, 2017). Furthermore, automobile buyers spend a significant amount of time on personal research (online blogs, YouTube, third party websites), visiting car dealerships, talking with their peers (family, friends, and co-workers), and being exposed to advertisements (print and media) before deciding to purchase (COX Automotive, 2019; van Rijnsoever et al., 2009). Although consumers consider all these sources, interactions

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with peers and visiting a car dealership significantly impact their decision to finalize the purchase (Cars.com, 2014; Dehdashti et al., 2018; Klein and Ford, 2003; van Rijnsoever et al., 2012).

Connected and autonomous vehicles (CAVs) utilize technology to accelerate, brake, steer and communicate with other on-road vehicles or infrastructure than humans in conventional vehicles. In the past decade, prolific research and industry-related efforts on the adoption CAVs vindicate the imminent addition of such vehicles either as an alternative travel mode or replacing/upgrading their existing conventional vehicle counterparts (Bansal and Kockelman, 2017; Bezai et al., 2021; Fagnant and Kockelman, 2015; Zhang et al., 2021). However, like every other technological innovation, CAVs also include several associated adoption barriers that might hinder widespread consumer acceptance (Bezai et al., 2021; Hulse et al., 2018; Milakis and Müller, 2021; Zhang et al., 2021). As per the most recent germane literature review (Bezai et al., 2021), principal concerns for CAV acceptance include low user acceptance, safety, and legislation. There is a dire need to shift the research focus toward consumers' acceptance of CAVs from their adoption (Hulse et al., 2018; Milakis and Müller, 2021). Also, research gaps exist in formulating methodologies focused on dealing with these anticipated uncertainties/barriers associated with CAVs (Li et al., 2019). Recent research highlights that consumers' CAV acceptance behavior will be affected due to their exposure to autonomous technology through social media, advertisements, and household/workplace interactions (Anania et al., 2018; Bansal and Kockelman, 2017; Poczter and Jankovic, 2013). Furthermore, educating consumers about CAVs while emphasizing their expected benefits (Haboucha et al., 2017; Liljamo et al., 2018) to increase consumers' trust in autonomous technology (Zhang et al., 2021) will surely boost their broader adoption rate. Electric vehicle adoption research also indicates the impact of consumers' environmental concerns on their intention to purchase electric vehicles (Wang et al., 2017).

To the best of our knowledge, a long-lasting gap exists in capturing the impact of pre-purchase information on CAV acceptance, mainly through an econometric framework. Furthermore, despite prolific studies highlighting the potential consumers' increased concerns about the anticipated barriers of CAVs (Bezai et al., 2021; Raj et al., 2020), little is known about such consumers' preferred source(s) of obtaining information about CAVs to minimize the risks and maximize the satisfaction associated with the post-purchase experience. In addition, past literature on conventional and electric vehicle purchases highlights the impact of potential consumers' concerns and multiple information channels on their vehicle purchase decision. Therefore, this paper seeks to answer the following questions based on random utility maximization theory, i.e., (1) *How to quantify the role of multiple information sources in disseminating pre-purchase information to potential CAV buyers. (2) How is this dissemination different for buyers who are skeptical of CAV than indifferent. (3) Do these sources complement or substitute one another or, in other words. (4) Do consumers rely on more than one information source? (5) What are the policy implications for stakeholders involved in CAV development and their market introduction to boosting the CAV acceptance rate.* Quantification of these questions will assist in shifting the focus from CAV adoption to their acceptance while assisting policymakers in resolving negative aspects of CAVs through information dissemination using appropriate channel(s).

The remainder of the paper is organized as follows. Section 2 discusses relevant literature on identifying CAV adoption barriers and exploring the impact of multiple information channels on vehicle purchase. Section 3 describes the data utilized, whereas Section 4 delineates the methodology, including mathematical formulation, followed by the results and discussions of this study in Section 5. Section 6 discusses key policy implications and section 7 concludes the

findings. Finally, section 8 highlights limitations and future research avenues of the study.

2. Literature Review

Numerous literature reviews exist to compile safety, travel behavior, consumer acceptance, and resistance-related impacts of CAVs (Bagloee et al., 2016; Becker and Axhausen, 2017; Bezai et al., 2021; Fagnant and Kockelman, 2015; Gkartzonikas and Gkritza, 2019; Raj et al., 2020; Soteropoulos et al., 2019; Zhang et al., 2021). Among the resistances, existing studies explored the liability, licensing, regulation, privacy/security, safety, cost, less customer acceptance, performance, and induced travel-related concerns of CAVs (Bagloee et al., 2016; Bansal et al., 2016; Bansal and Kockelman, 2017; Buckley et al., 2018; Fagnant and Kockelman, 2015; Gkartzonikas and Gkritza, 2019; Gurney, 2013; Haboucha et al., 2017; König and Neumayr, 2017; Kyriakidis et al., 2015; Li et al., 2019; Raj et al., 2020; Shabanpour et al., 2018). Table 1 provides a brief overview of studies considering barriers associated with CAVs and the impact of multiple information sources on vehicle purchase.

For conventional automobiles, research in considering different sources of information dates back to the 90s when Moorthy (Moorthy et al., 1997) showed that print media and car dealers dominated the primary source of information for purchasing new cars. Later, Morton et al. (2001) used a linear regression model to show that considering the internet before visiting car dealers resulted in better automobile purchases (getting more purchase discounts). Klein and Ford (2003) also used regression to indicate that the internet substituted car dealers. Ratchford et al. (2003) found that internet use saved significant time searching automobile purchase-related information. Furthermore, Ratchford et al. (2007) used the binary probit model to find that internet use decreased time negotiating with the salesperson at car dealerships. van Rijnsoever et al. (2009) utilized structural equation modeling to find a positive relationship between attitude towards vehicle's performance and using media and the internet for information. Finally, van Rijnsoever et al. (2012) found a positive association between purchase involvement and considering the increased number of information sources. Among socioeconomic characteristics, age was related positively to the number of sources considered before purchasing the automobile. In contrast, educational attainment was associated negatively (Klein and Ford, 2003). Furthermore, age was positively related to personal research but negatively to the internet and peer-to-peer communication (van Rijnsoever et al., 2012). Older individuals relied heavily on car dealerships (Parment, 2013). Young, low-income earners and educated buyers preferred the internet for their automobile purchases (Dehdashti et al., 2018; Kim and Ratchford, 2012a). Attitudes towards the vehicle's performance were positively related to searching mass media and internet information (van Rijnsoever et al., 2009).

From the CAV acceptance point of view, to the best of our knowledge, past efforts in capturing the impact of different information sources on individuals concerned about the negative aspects of CAVs are limited (Anania et al., 2018; Ghasri and Vij, 2021; Zhu et al., 2020). From the adoption perspective, Zhu et al. (Zhu et al., 2020) captured the impact of mass and social media on CAV adoption using structural equation modeling on a survey of 355 students in China. The results showed that media information directly impacts individuals' perception of CAV, indirectly influencing individuals' decisions to adopt CAVs. Anania et al. (2018) used reliability and inferential analysis on two surveys of Americans and Indians to capture the impact of positive and negative information on their willingness to ride in CAVs. The results showed a direct relationship between positive (negative) information and willingness (unwillingness) to ride in a CAV. Using the latent class choice model on 862 Australian residents, Ghasri and Vij (2021) captured the impact of social

and mass media on CAV adoption. The authors found that social media sentiment has the highest effect on AV consideration (90% of the sample).

From the conventional vehicle literature, car dealers, mass media, and the internet were among the most used and trusted sources of information for finalizing the decision to purchase. To the best of our knowledge, such literature is non-existent in considering the importance of pre-purchase-related information obtained from multiple sources (peer interactions, visiting the nearest car dealership, and personal research) for CAVs.

2.1 Contribution

This study attempts to answer the policy-related question: *"How to quantify the role of multiple information sources in the effective dissemination of pre-purchase information to the potential CAV buyers, especially those who are skeptical of autonomous technology."* This study is distinct from existing CAV adoption/acceptance studies in four important ways. *First*, we clearly consider four different information sources (peers, car dealers, media advertisements, and personal research over the internet) and their impact on disseminating CAV-related pre-purchase information to the demographics of potential consumers. This study is different from existing studies exploring the impact of word of mouth (Anania et al., 2018; Ghasri and Vij, 2021) on CAV adoption. These studies explore the influence of positive and negative media information (Anania et al., 2018; Ghasri and Vij, 2021) and word of mouth from CAV adopters (Ghasri and Vij, 2021) on individual's intention to adopt CAVs. Unlike these studies, we simultaneously capture the impact of information disseminated from the four sources on individuals' acceptance of CAVs, i.e., finalizing the decision to purchase, not adopt. In other words, we consider the consumers who depend on different information channels on whether to accept a CAV as their next vehicle or not. The findings will assist potential consumers in utilizing these channels to complement their CAV purchasing decision. Such quantification will allow car dealerships and technology companies to communicate information effectively with potential consumers and increase CAV sales. Furthermore, the exploration of this kind will provide future avenues of research in investigating the impact of such a relationship on the price and overall market penetration of CAVs. *Second*, we narrow down this impact of CAV information obtained from considered sources on potential consumers who are skeptical and indifferent towards the negative impacts of CAVs. To the best of our knowledge, existing studies in this context are non-existent. This analysis is expected to guide policymakers and planners in framing policies to deal with resistances associated with CAVs. *Third*, we attempt to identify the demographics and lifestyle of omnichannel consumers, i.e., who rely on more than one information source. The findings are expected to increase the outreach of CAV-related information disseminated through a single channel to a larger audience. *Fourth*, complementing the third contribution, we explore the interrelations (complementary/substitution effects) among all four information sources, which will assist stakeholders involved in CAV sales to increase the effectiveness of a particular information source by coalescing it with the correlated source(s).

Table 1. Literature review: past literature on barriers associated with CAV adoption and impact of information sources on vehicle purchase

Source	Data	Focus area	Method	Key barriers/findings
			CAV barriers	
Bagloee et al. (2016)	Literature synthesis	Hurdles associated with CAVs	Navigation model	Customer acceptance, ethical issues, licensing standards and cybersecurity
Bansal et al. (2016)	347 American residents	CAV adoption	Ordered probit model	System failure, cybersecurity, privacy, affordability, crash liability and learning curve associated with CAVs.
Berliner et al. (2019)	2,261 American residents	CAV adoption	Ordered logit model	Safety as compared to conventional vehicles
Bezai et al. (2021)	Literature synthesis	Barriers to CAV adoption	Taxonomy approach	Principal clusters of barriers including, user acceptance/reaction, safety, and regulations. The user acceptance cluster included the public's perception of CAV usage and the cost.
Buckley et al. (2018)	68 American drivers	CAV adoption	Driver simulator test	Cybersecurity and privacy.
Fagnant and Kockelman (2015)	Literature synthesis	Barriers to CAV adoption	Literature review	High initial costs of CAVs, disparate state legislations and standards, uncertainty in AV crash liability, security and privacy concerns
Haboucha et al. (2017)	731 American and Israeli residents	CAV adoption	Mixed logit model	Environmental concern, enjoying the driving, and cost
Howard and Dai (2014)	107 American residents	CAV adoption	Logit model	Cost, privacy, safety, and lack of control
Hulse et al. (2018)	925 UK residents	CAV adoption	Multinomial logit model	Cybersecurity, lack of knowledge about safety-related capabilities, and privacy
Jiang et al. (2019)	1,002 Japanese residents	CAV adoption	Mixed logit model	Purchase and maintenance costs
König and Neumayr (2017)	489 residents from 33 countries	Barriers to CAV adoption	Descriptive statistics	Cybersecurity, lack of control safety, and crash liability
Kyriakidis et al. (2015)	5,000 residents of 109 countries	CAV acceptance	Descriptive statistics and correlation analysis	Legal, cybersecurity, and safety issues
Liljamo et al. (2018)	2,036 Finnish residents	CAV acceptance	Cross-tabulation	Traffic safety and technical reliability (driver takes over under emergencies).
Raj et al. (2020)	Literature synthesis and expert opinions	Barriers to CAV adoption	Multicriteria decision-making technique	Crash liability, low consumer acceptance, decreased employment opportunities, insufficient standards/regulations, high manufacturing costs, privacy, and cybersecurity.
Sanbonmatsu et al. (2018)	114 American residents	CAV adoption	Correlation analysis	Unfamiliarity and low trust in autonomous technology
Shabanpour et al. (2018)	1,253 American residents	CAV adoption	Best-worst model	Cost, unreliable CAV operation under unexpected traffic conditions, and crash liability

Table 1 continued

Source	Data	Focus area	Method	Key barriers/findings
		Information channels and vehicle purchase		
Anania et al. (2018)	Multiple surveys in US and India (N= 64 and 992)	Impact of information on CAV adoption	Reliability and inferential analysis	Positive and negative information related to CAVs increased and decreased individuals' willingness to ride in CAV.
Dahiya and Gayatri (2018)	784 Indian residents	Advertisements and automobile purchase	Multiple significance tests	75% of individuals relied on digital media channels (website and social media) while deciding to purchase a car.
Dehdashti et al. (2018)	3,473 car buyers in the US (1990-2012)	Information sources and car purchase	Latent class model	Education and low income were positively related to the internet-based sources
Ghasri and Vij (2021)	862 Australian residents	Media and CAV adoption	Latent class choice model	Social media sentiment has the highest effect on AV consideration (90% of the sample).
Klein and Ford (2003)	171 and 168 car buyers and shoppers	Consumer's information search behavior	Multiple linear regression model	Internet substituted traditional sources of information, especially car dealerships. Furthermore, age and educational attainment were related positively and negatively with the number of sources considered before purchasing the automobile.
Moorthy et al. (1997)	51 American residents (new car buyers in 1993)	Consumer's information search behavior	Multiplicative search model	Print media and car dealers dominated the primary source of information for the new cars
Morton et al. (2001)	324,936 vehicle purchases (1999-2000)	Effect of the internet on the vehicle purchase	Multiple linear regression model	Internet use results in better automobile buys (discounts purchase costs by 1.5-2%).
Ratchford et al. (2003)	Two surveys of 901 and 886 car buyers in the US.	Consumer's information search behavior	Multiple linear regression model	Internet saves significant time in searching for automobile purchase-related information. Younger and educated are more likely to use the internet over other sources.
Ratchford et al. (2007)	Three surveys of 843, 822 and 705 car buyers in US.	Consumer's information search behavior	Multiple linear regression and binary probit model	Internet decreases significant time spent interacting with the salesman in the nearest dealership.
van Rijnsoever et al. (2009)	1,392 Car owning households in Netherlands	Information sources and car purchase	Structural equation modeling	Attitudes towards the vehicle's performance were positively related to using mass media and the internet for information.
Kim and Ratchford (2012)	1,665 car buyers in the US (1999-2005)	Information sources and car purchase	Multivariate ordered logit	Young and educated buyers prefer the internet for their automobile purchases. Younger, single, inexperienced and inexpensive car buyers rely on friends for the information.
van Rijnsoever et al. (2012)	1,261 Netherland residents	Information sources and car purchase	Structural equation modeling	Age is positively related to personal research but negatively to the internet and peer-to-peer communication. Purchase involvement is related to consulting an increased number of information sources.
Jang et al. (2017)	3,198 car buyers in US (2002-12)	Information sources and car purchase	Tobit model with latent classes	A positive effect of being married on consulting offline dealerships and the positive association between online sources and their offline counterparts
Zhu et al. (2020)	355 Chinese students	Media and CAV adoption	Structural equation modeling	Media information, directly and indirectly, impact individuals' perception of CAV and their decision to adopt CAVs.
This study	*4,448 American residents*	*Adoption barriers, information sources, and CAV acceptance*	*Latent class and multivariate ordered logit*	*Among concerned individuals, the elderly, females, and residents willing to pay more than $30k for a conventional car are more likely to rely on car dealers when purchasing a CAV.*

3. Survey design and data processing

A statewide online survey was disseminated in Tennessee from March 2020 to May 2020 using the web-based survey tool, Qualtrics. To select a representative sample size, we targeted to engage 4,600 residents, assuming a 95% confidence level and $\pm 2\%$ level for precision in Cochran's formula (Cochran, 1977). The survey's main objective was to capture Tennesseans' perceptions of anticipated capabilities and negative aspects of level-4 CAVs. The survey included different blocks, including questions about socioeconomics (individual and household-level), vehicle ownership, work, daily commute, tech-savvy lifestyle, established peer network, and CAV-related attributes. The main highlight of the survey was to capture the anticipated impact of residents' decision to adopt CAVs on their peer social networks and their perception of the anticipated negative impacts of CAVs.

Institutional Review Board (IRB) approved the survey in an expedited track. The survey was disseminated in three different channels, i.e., Amazon Mechanical Turk (MTurk), Facebook advertisements and emails to educational institutes, and a Panel from a leading marketing research company. Incentives were offered in the first two channels but not in the third as we paid the company for each collected response. We received about 5,646 responses in total, and after removing the incomplete responses, the sample was reduced to 4,602. We removed all responses with a response time of less than 40% of the median completion time for keeping the quality responses. The sample was finally reduced to 4,448 responses, where 71%, 19%, and 9% of participants were hired from the market research company's panel, MTurk, and educational institutes cum social media, respectively. It is worth mentioning that the survey was conducted during the initial stay-at-home lockdown in the State of Tennessee due to the COVID-19 pandemic (Asquith, 2020); hence it might have affected the nature of responses when compared to pre-pandemic times. In the upcoming subsections, we describe survey results in terms of the dependent variable, summary statistics of covariates, and respondents' concern toward negative aspects of CAVs.

3.1 Dependent variable: Residents' dependence on different CAV information sources

As part of the survey, we asked residents about their perception of four different information sources before finalizing their decision to purchase a CAV., i.e., a friend/colleague, media advertisements, nearest car dealer, and personal research (Fig. 1). The survey questions were asked on a 7-level Likert scale (Very Unreliable to Very Reliable). However, for modeling purposes, we transformed the scale into a 3-level scale. Interestingly, respondents preferred personal research and peer networks over car dealers and media advertisements. Respondents preferring personal research are in line with Gooogle's survey, suggesting that YoutTube influenced 70% of auto shoppers (Google, 2018). Respondents had mixed opinions about media advertisements as the majority of participants had neutral responses towards their dependence on such sources.

3.2 Summary statistics

Table 2 delineates the descriptive statistics of the sample. The variables are segregated based on their nature and their respective question block in the survey. For some variables in Table 2, the survey question had different levels. However, we combined some levels for analysis to ensure an equal or comparable percentage of responses in each level. Most respondents are aged 18 to 35 years, female, college graduates, White, married, and earn between $35,000 and $100,000. When comparing these attributes with Tennessee's population (Fig. 2) as per the 2018 American

Community Survey (ACS) (Manson et al., 2019), the sample overrepresented[1] married, wealthy (annual income over $75,000), and highly qualified (college graduate or higher). In contrast, the sample underrepresents low-income (annual income less than $25,000) and less qualified (high school or below) residents. Since the survey was conducted entirely online, the sample might be skewed towards the residents with internet access, as only 78% of households in Tennessee have access to broadband internet. The majority of residents are located in suburban areas and include two or fewer members in their household. About 87% of respondents do not have any disability undermining their driving ability, and 57% plan to buy or sell a car in the next three years. Among vehicle ownership characteristics, the majority of households (respondent's) purchase a vehicle every five to ten years, are willing to pay less than $15,000 for their next car, own about two cars, purchased at most one new car and at least two used cars in the last ten years and drive more than 5,000 miles annually.

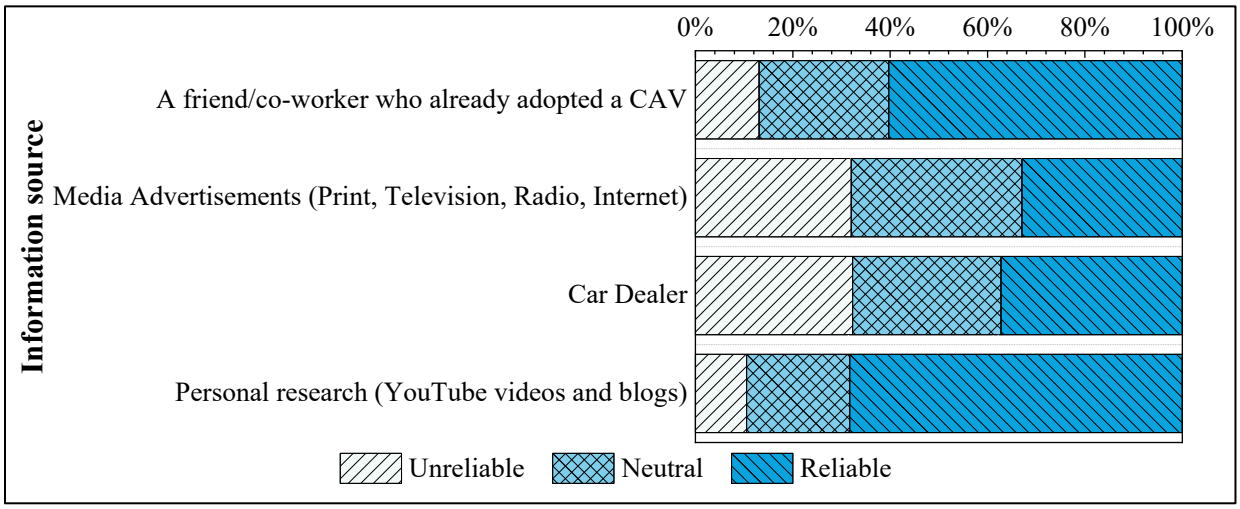


Fig. 1. Dependent variable: Respondent's dependence on different information sources when deciding to purchase a CAV (N = 4,448)

The majority of respondents never work from home, have a flexible work schedule, use private cars for their daily commute, and never use other forms of transportation for their daily commute. Furthermore, in terms of tech-savvy lifestyle-related attributes, most residents own a smartphone, listen to radio and watch TV daily, use GPS navigations services at least once a week, and never use smart home services. Despite the familiarity with CAVs, the majority of respondents are willing to pay less than $2,500 extra for installing autonomous technology in their conventional car, not willing to pay anything to maintain the technology, and not interested in using any of the CAV-based travel modes (privately owned, carpooled, public transport and ride-hailing service). On average, residents have 22 peers in their social network, with at least two peers enthusiastic about cars. Such residents communicate at least two to three times a week with these peers.

[1] Considered overrepresentation or underrepresentation if difference between population and sample percentages is more than ±5%

Table 2. Descriptive statistics of the sample (N=4,448)

Variable		Percentage	Variable		Percentage
Socioeconomic variables					
Age	18 to 35 years	35%	Ethnicity	White	78%
	35 to 54 years	31%		African American	14%
	more than 54 years	34%		Others	8%
Marital Status	Single	28%	Annual income	Less than $35,000	35%
	Married	58%		$35,000-$75,000	33%
	Separated, divorced, widowed	14%		more than $75,000	32%
Educational attainment	High school or below	21%	Gender	1 = Male	42%
	Some College	27%		0 = Female	58%
	College Graduate	30%			
	Master's or professional degree	22%			
Household-related variables					
Household locality	Urban	27%	Household members	Two or less	55%
	Semi-urban	41%		Three or more	45%
	Rural	32%			
Vehicle ownership related variables					
Household Cars	One and zero	37%	Conventional car purchase frequency	Once a year to every 2 to 3 years	22%
	Two	40%		Once every 5 to 10 years	66%
	Three or more	23%		Once every 15 to 20 years	12%
Physical disability undermining driving	1 = Yes	13%	Plans to sell/buy a car in next three years	1 = Yes	57%
	0 = No	87%		0 = No	43%
New cars in the household (last ten years)	One and zero	37%	Used cars in the household (last ten years)	Zero	32%
	Two	32%		One	28%
	Three or more	31%		Two or more	40%
Willingness to pay for a conventional car	less than $15,000	40%	Driving mileage (annual)	less than 5,000	46%
	$15,000-$30,000	35%		more than 5,000	54%
	more than $30,000	25%			
Work and daily commute-related variables					
Remote work frequency	Frequent*	41%	Work schedule flexibility	Flexible	45%
	Infrequent*	12%		Somewhat flexible	25%
	Never	46%		Inflexible	30%
Private car usage frequency	Frequent*	84%	Public transport usage frequency	Frequent*	7%
	Infrequent*	5%		Infrequent*	5%
	Never	11%		Never	88%

Table 2 Continued

Variable		Percentage	Variable		Percentage
Ridesharing service usage frequency	Frequent*	6%	Ride hailing service usage frequency	Frequent*	5%
	Infrequent*	6%		Infrequent*	6%
	Never	89%		Never	89%
Tech-savviness related variables					
Radio listening frequency	Frequent*	56%	TV watching frequency	Frequent*	82%
	Infrequent*	39%		Infrequent*	16%
	Never	4%		Never	2%
Smart home usage (Amazon Alexa) frequency	Frequent*	25%	GPS navigation usage frequency	Frequent*	19%
	Infrequent*	19%		Infrequent*	67%
	Never	56%		Never	14%
Own a Smartphone	1 = Yes	95%			
	0 = No	5%			
CAV related variables					
Willingness to pay for installing autonomous technology in existing conventional car	less than $2,500	45%	Willingness to pay for annual maintenance of AV technology	$0	41%
	$2,500 to $7,500	30%		$0 to $300	28%
	more than $7,500	25%		more than $300	31%
Familiar with AV technology	1 = Yes	79%	Willing to pay for automated service station trips in CAVs	1 = Yes	39%
	0 = No	21%		0 = No	61%
Interested to privately own a CAV.	1 = Yes	41%	Interested to use ride hailing CAV with a backup driver	1 = Yes	32%
	0 = No	59%		0 = No	68%
Interested to use ride hailing CAV with no backup driver	1 = Yes	22%	Interested to use Autonomous Public Transport service	1 = Yes	28%
	0 = No	78%		0 = No	72%
Interested to carpool/share a CAV	1 = Yes	25%	Involved in vehicle crashes in the past	1 = Yes	59%
	0 = No	75%		0 = No	41%
Peer network related variables					
Peers knowledgeable about cars	Zero	28%	Peers' communication frequency	Frequent**	55%
	1 to 2 peers	30%		Sometimes**	21%
	more than 2 peers	42%		Infrequent**	24%
Number of established peers	Mean	21.59	*Survey channel related variables*		
	Standard Deviation	27.18	Survey distribution channel	Amazon Mechanical Turk	19%
	Min/Max	0/100		Social media and institutes	9%
				Market research company	71%

* Frequent: everyday; Infrequent: (once a week to a year).

** Frequent: 2-3 times a week to daily; Sometimes: every couple of weeks to month; Infrequent: once per month to every few months

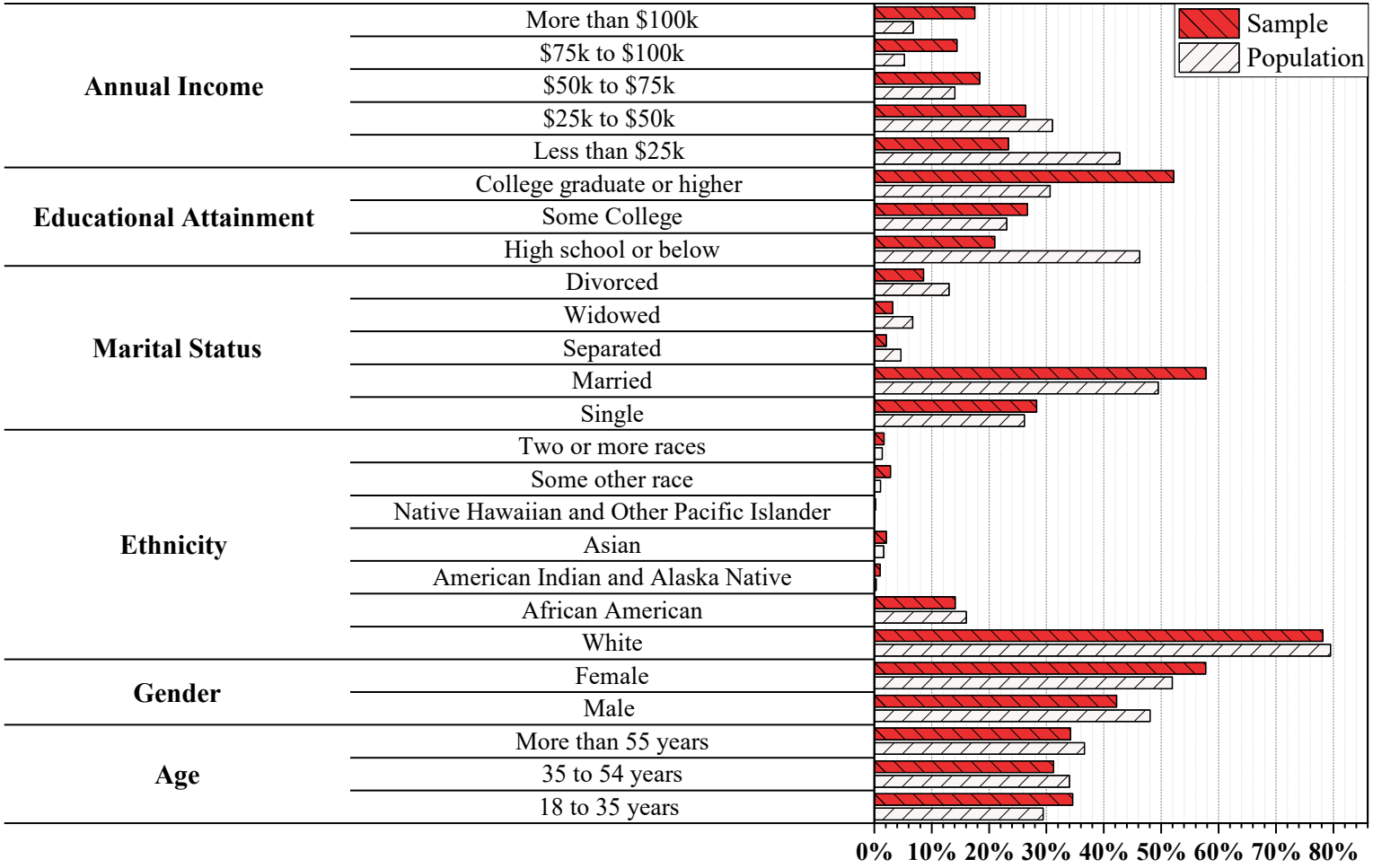


Fig. 2. Demographics comparison: sample and population.

3.3 Respondents' concern towards negative aspects of CAVs

In terms of capturing residents' concern about the anticipated negative impacts of CAVs, we considered eight different impacts being consistent with the existing literature (Buckley et al., 2018; König and Neumayr, 2017; Lu et al., 2017; Sharma and Mishra, 2020; Talebian and Mishra, 2018) covering the operation, privacy, safety and maintenance related indicators. Initially, indicators were measured on a 7-level Likert scale (Very Unimportant to Very Important). However, for modeling purposes, we transformed them to a 5-level Likert scale. Indicators with their modeling notation (C1 to C8) and percentage distribution of responses are portrayed in Fig. 3. Overall, Respondents were highly concerned about all eight negative impacts of CAVs, followed by indifference which is in line with existing literature (Kim et al., 2022a). The majority of respondents were highly concerned about CAV system failure (C3 and C4), followed by the safety of CAVs when compared to conventional vehicles (C7) and the possibility of taking over driving control when CAV fails (C2). Comparatively, residents were least concerned about the possibility of high maintenance costs for CAVs compared to conventional cars (C8).

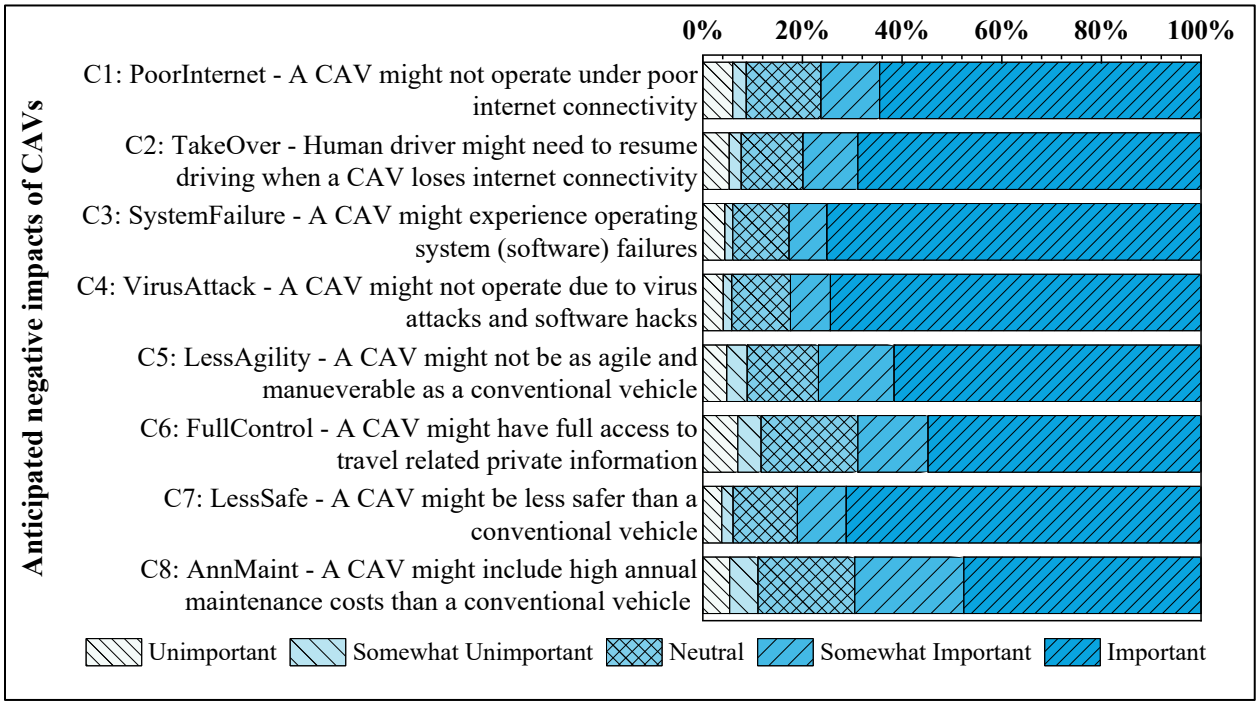


Fig. 3. Respondents' concern towards the negative aspects of CAVs (N= 4,448)

4. Modeling approach

To capture the residents' dependence on different information sources, we first perform a latent class analysis to segregate the sample based on residents' concerns about the negative impacts of CAVs. Then we utilized a multivariate ordered logit model to capture the dependence on four different information sources when purchasing a car for the segregated subsamples. The modeling framework is shown in Fig. 4, along with the modeling notations of variables, wherever applicable. In the upcoming sections, we give a brief overview of both models' mathematical formulation and estimation.

4.1 Latent class analysis

We utilize latent class analysis to segregate the sample based on residents' concerns about the negative impacts of CAVs. As discussed in the data section, we included eight Likert scale-based attitudinal indicators to capture individuals' concerns about the negative impacts of CAVs (C1-C8). As per latent class analysis, a discrete latent variable can quantify the relationship among these attitudinal indicators making the relationship insignificant and conditional on latent class variables (Kemperman and Timmermans, 2009). This section discusses the mathematical framework for latent class analysis based on existing literature (McCutcheon, 1987).

Let y_{nil} denotes the lth response of individual n for attitudinal indicator variable i (Also known as manifest variables). In our case, l represents available response outcomes for each indicator variable i (5-level Likert scale, i.e., $l \in L = \{1,2,3,4,5\}$). The dataset includes eight indicator variables, i.e.,$i \in I = \{C1, C2, C3, C4, C5, C6, C7, C8\}$ delineated in Fig. 3. Also, $y_{nil} = 1$ if individual n gives l^{th} response to the ith indicator variable and $y_{nil} = 0$ otherwise. Let θ_{irl} represents the class conditional probability that an individual responding to the lth outcome for variable i will be in class r = 1,2,3….,R where R is the total number of classes fixed before estimation based on a model fit criterion or theoretical reasoning such that $\sum_{l=1}^{L} \theta_{irl} = 1$. Let $p_r(x_n; \gamma)$ be the R mixing proportions indicating the weighted sum of all the classes such that, $\sum_{r=1}^{R} p_r(x_n; \gamma) = 1$ where x_n represents the observed covariates for respondent n and γ denote the coefficient vector explaining the structural relationship for covariates in each class r.

The probability that a respondent n completes a particular set of l responses on all indicator variables belonging to class r is given by

$$f(y_n; \theta_r) = \prod_{i=1}^{I} \prod_{l=1}^{L} (\theta_{irl})^{y_{nil}} \tag{1}$$

Where I and L are sets of all indicator variables and response outcomes for each indicator variable. The probability function after accounting for the weighted sum across classes and covariates is given by

$$P(y_n|\theta, p) = \sum_{r=1}^{R} p_r(x_n; \gamma) \prod_{i=1}^{I} \prod_{l=1}^{L} (\theta_{irl})^{y_{nil}} \tag{2}$$

The log-likelihood function is given by

$$\ln L = \sum_{n=1}^{N} ln \sum_{r=1}^{R} p_r(x_n; \gamma) \prod_{i=1}^{I} \prod_{l=1}^{L} (\theta_{irl})^{y_{nil}} \tag{3}$$

The class memberships θ_{irl} and parameters γ can then be obtained after maximizing the log-likelihood function using the expectation-maximization algorithm (Dempster et al., 1977) while using the multinomial logit link function to cater for the effect of covariates on the priors, i.e., $p_r(x_i; \gamma)$ (Agresti, 2003). We utilized "poLCA" package in R (Linzer and Lewis, 2011) to formulate and estimate latent class analysis parameters.

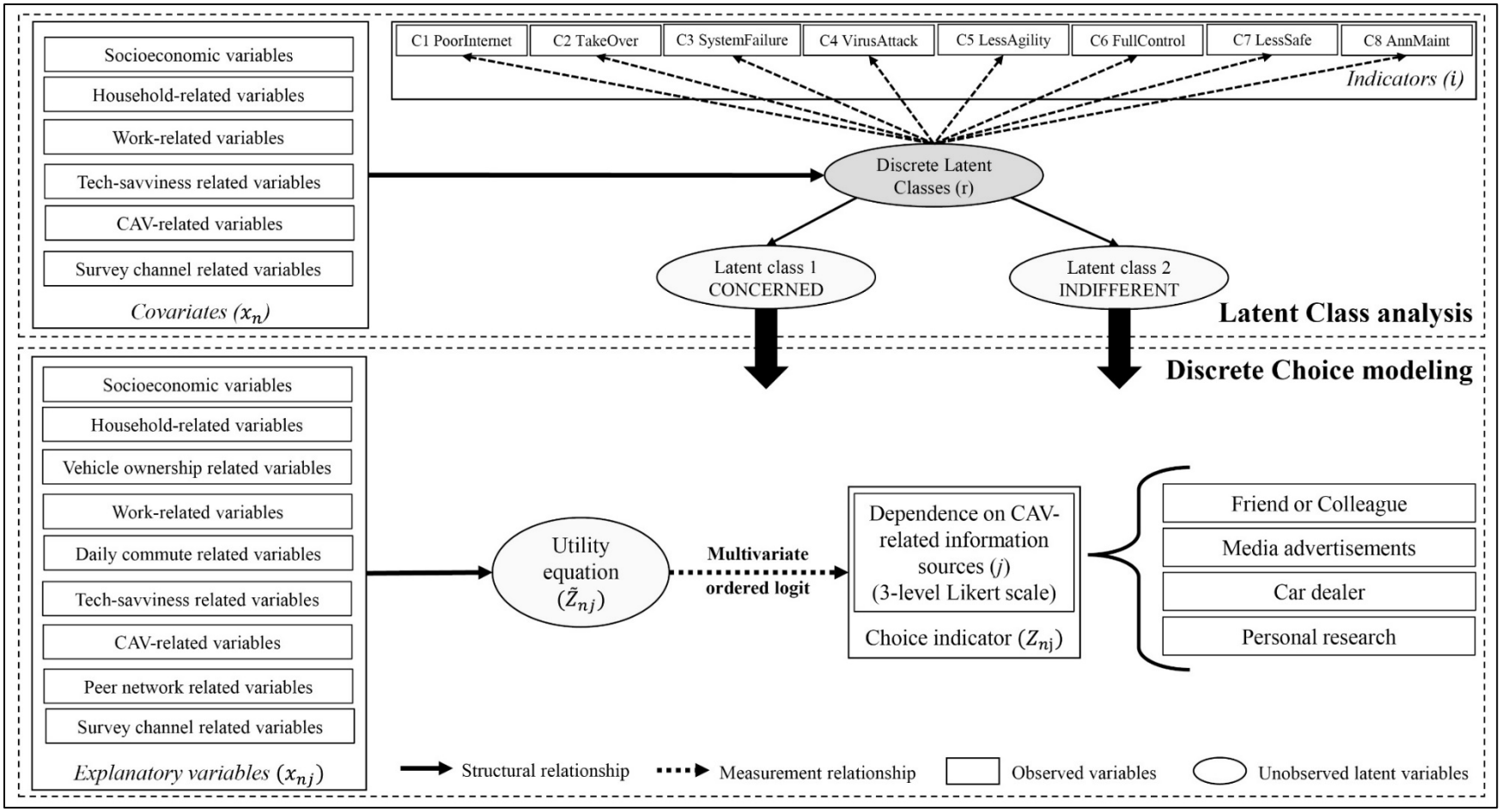


Fig. 4. Modelling approach

4.2 Multivariate ordered logit

Individuals depend on more than one information source before finalizing their decision to purchase an automobile (COX Automotive, 2019; van Rijnsoever et al., 2009). Hence, we utilize a multivariate ordered logit model, an extension of their univariate counterparts (Hirk et al., 2020), to capture the dependence on four different information sources (including their error correlations) when deciding to adopt a CAV for identified latent classes.

Let Z_{nrj} denote ordinal response of individual n belonging to class n for outcome variable j where $j \in J =$ {Friend/colleague, media advertisements, car dealer, personal research}. Let x_{nrj} be the covariates (Table 2) associated with individual n belonging to class r and outcome j. Such model can be formulated using cumulative link models where the observed ordinal outcome Z_{nrj} can be represented by a continuous latent variable $\tilde{Z}_{nrj}$ (Hirk et al., 2020) and both are related as:

$$Z_{nrj} = o_{nrj} <=> \pi_{rj,o_{nrj}-1} < \tilde{Z}_{nrj} \leq \pi_{rj,o_{nrj}}, o_{nrj} \in \{1, \dots, O\} \quad (4)$$

Where o_{nrj} is the Likert scale response given by respondent n, belonging to class r, to outcome j in O-level Likert scale ($O = 3$ in our case as per Figure 1) and π_{rj} is a vector of threshold parameters for outcome variable j and class r. Next, the latent variable $\tilde{Z}_{nrj}$ can then be related linearly with the vector of exogenous variables x_{nrj} as:

$$\tilde{Z}_{nrj} = \alpha_{jr0} + \alpha_{jr} x_{nrj}^{\tau} + \varepsilon_{nrj} \quad (5)$$

Where, α_{jr0} and α_{jr} are the intercept and vector of unknown regression coefficients for outcome variable j and class r. We assumed each individual n as independent of other individuals. To account for the heterogeneity among the outcomes, the unknown parameters, including thresholds π_{rj} and coefficients α_{rj} were allowed to vary over outcome variables ($j \in J$). To exploit the log odds interpretation for the estimated parameters, we assumed a multivariate logistic distribution for the vector of error terms ε_{nrj} with t copula (O'Brien and Dunson, 2004) and univariate logistic margins. The model can then be estimated using the composite likelihood approach (Varin et al., 2011). For more details, readers are referred to (Hirk et al., 2020) as we used the package "mvord" in R -programming to formulate and estimate the model.

5. Results and discussions

5.1 Latent class analysis

We selected an optimal number of classes based on the model interpretation, adequate class size (> 10%), the goodness of fit measures (log-likelihood), and statistical measures like Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). All measures, especially BIC, show improvement with an increase in the number of classes. We chose the 2-class model over the optimal 7-class model because the former (Table 3) provided better interpretability in terms of distinct classes based on the item response probabilities (Table 4). It is not uncommon to use a different number of classes based on their interpretability (Eldeeb and Mohamed, 2020; Ferguson et al., 2018; Weller et al., 2020).

Based on individuals' eight different attitudes towards the negative aspects associated with CAVs (C1-C8), we identified two different segments of residents, i.e., "Concerned" and "Indifferent". In the rest of the sections, we use these terms to refer to latent classes. In latent class Concerned, maximum item response probability for each indicator variable falls in Likert scale level "Important" (bold values in Table 4). Based on these values, it can be inferred as residents

conforming to this class are highly concerned about the negative aspects of CAVs. On the other hand, maximum probabilities for each indicator in latent class Indifferent fall in the "Neutral" outcome level, which implies that such residents are unconcerned or nonchalant towards the negative aspects of CAVs. As per latent class size, about 77% of respondents are Concerned, whereas the remaining are Indifferent, which is in line with Kim et al. (2022).

Table 3. Latent class selection criteria (N=4,448)

Model	Number of parameters	Log-Likelihood (LL)	AIC	BIC	Entropy
1-Class	32	-37,737.05	75,538.09	75,742.90	8.48
2-Class	65	-31,467.83	63,065.65	63,481.67	7.10
3-Class	98	-29,820.97	59,837.95	60,465.17	6.74
4-Class	131	-28,679.23	57,620.45	58,458.88	6.49
5-Class	164	-28,282.55	56,893.10	57,942.74	6.39
6-Class	197	-27,961.73	56,317.45	57,578.29	6.31
7-Class	230	-27,775.35	56,010.70	57,482.75	6.27
8-Class	263	-27,690.13	55,906.27	57,589.52	6.25

Table 4. Conditional Item response probabilities (N=4,448)

Likert scale level	Latent Class		Latent Class	
	Concerned	Indifferent	Concerned	Indifferent
Cluster size	76.68%	23.32%	76.68%	23.32%
	C1: PoorInternet		*C5: LessAgility*	
Unimportant	0.021	0.191	0.014	0.162
Somewhat Unimportant	0.010	0.081	0.025	0.093
Neutral	0.054	**0.468**	0.047	**0.463**
Somewhat Important	0.104	0.164	0.141	0.189
Important	**0.811**	0.095	0.773	0.093
	C2: TakeOver		*C6: FullControl*	
Unimportant	0.010	0.193	0.039	0.172
Somewhat Unimportant	0.004	0.090	0.036	0.082
Neutral	0.027	**0.449**	0.113	**0.463**
Somewhat Important	0.090	0.180	0.133	0.168
Important	**0.869**	0.088	0.679	0.114
	C3: SystemFailure		*C7: LessSafe*	
Unimportant	0.001	0.186	0.007	0.145
Somewhat Unimportant	0.000	0.070	0.005	0.081
Neutral	0.010	**0.456**	0.033	**0.450**
Somewhat Important	0.043	0.188	0.072	0.185
Important	**0.947**	0.100	0.884	0.140
	C4: VirusAttack		*C8: AnnMaint*	
Unimportant	0.003	0.166	0.022	0.163
Somewhat Unimportant	0.005	0.060	0.046	0.092
Neutral	0.015	**0.460**	0.121	**0.439**
Somewhat Important	0.048	0.187	0.231	0.178
Important	0.930	0.127	0.581	0.128

***Bold** values indicate the maximum probability

Goodness of fit measures: Log-likelihood = -25,661; AIC = 51,484; BIC = 52,003

In addition to the item response probabilities, we also added the effect of covariates on class membership. We included socioeconomic, household, work, tech-savviness, CAV, and survey channel-related variables in the model to capture the impact of socioeconomics, work patterns, tech-savvy lifestyle, and CAV-related attributes on the class membership. The results are delineated in Table 5. The first column includes explanatory variables and their estimated level (s) and base level. For the estimation, the latent class Concerned, with a higher cluster size (N=3,417),

was kept as the reference, and we included only statistically significant estimates (p-value<0.10) in the final model. For the interpretation, the intercept coefficient can be inferred as ceteris paribus; residents are more likely to belong to the Indifferent latent class when compared to the Concerned latent class.

To simplify the class membership interpretation, we also estimated the relative risk ratio (RRR) as the exponential of the estimated coefficients as class memberships are related to covariates through a multinomial logit link function. A RRR value less than 1 indicates that the covariate belongs to the base class Concerned, whereas a value more than 1indicates that the covariate belongs to the latent class Indifferent (Vermunt and Magidson, 2005). The class membership RRR estimates for covariates are shown in Fig. 5. Individuals **aged over 35 years** are more likely to be concerned about the negative aspects of CAVs. This finding aligns with existing studies reflecting the higher interest of the younger population in CAVs (Liu et al., 2019; Robertson et al., 2017). Compared to females, **males** are more likely to be indifferent to negative aspects of CAVs consistent with (Kyriakidis et al., 2015). Compared to ethnicities other than White, **African Americans** are more likely to be Indifferent, which can be due to the higher affinity of such individuals towards using CAVs (Kassens-Noor et al., 2020). **College graduates** are more likely to be Concerned than less qualified individuals, which can be due to their increased awareness of the privacy concerns associated with technology (Acheampong and Cugurullo, 2019; Saeed et al., 2020).

Interestingly, individuals willing to pay higher (more than $7,500) towards installing AV technology are more likely to be indifferent. In contrast, individuals willing to pay higher for the maintenance of AV technology are more likely to be Concerned. Individuals **familiar with CAVs**, involved in road accidents in the past, interested in owning a CAV, and annual mileage over 5000 miles are more likely to be Concerned. This can be attributed to their increased interest to use CAVs (in line with (Saeed et al., 2020). Finally, **physically challenged individuals** are more likely to be indifferent. Such individuals are more likely to use CAVs to complete their mobility requirements, ignoring the negative aspects (Milakis and van Wee, 2020).

5.2 Multivariate ordered logit: class 1 and class 2

The modeling results for multivariate ordered logit, capturing Concerned and Indifferent individuals' dependence on different information channels, are delineated in Table 6. Consistent with latent class analysis, we included only significant variables (Column 1). For the interpretation, a positive (negative) sign of the coefficients indicates a high likelihood associated with the reliability (unreliability) of a particular information source. In contrast, the magnitude of the coefficient indicates the intensity of the estimates. For instance, the intercept coefficient for Indifferent latent class and friend/colleague information sources can be inferred as ceteris paribus; Indifferent individuals are more likely to depend on their friends or colleagues to collect CAV-related information while deciding to adopt one.

Compared to individuals aged between 18 to 35 years, Concerned individuals **aged more than 55 years** are more likely depend on car dealers for CAV-related information. These findings are consistent with existing literature as baby boomers, and older individuals are more likely to prefer car dealer consultations and in-store service (Dehdashti et al., 2018; Lambert-Pandraud et al., 2005; Parment, 2013; Ratchford et al., 2003, 2007). Indifferent individuals **aged 35 to 55 years** are more likely to find advertisements reliable.

Table 5. Class membership functions of latent class model (N: Class 1- Concerned = 3,417, Class 2- Indifferent = 1,031)

Variable	Indifferent (base class: Concerned) Coefficient Significance
Intercept	0.683***
Socioeconomic variables	
Age (base: 18 to 35 years)	
35 to 54 years	-0.411***
more than 54 years	-0.787***
Gender (base: Female)	
Male	0.461***
Ethnicity (base: others)	
African American	0.313*
Marital status (base: Separated, divorced, widowed)	
Single	0.177#
Educational attainment (base: High school or below)	
College Graduate	-0.262**
Annual income (base: <$35,000)	
$35,000-$75,000	-0.301***
Household-related variables	
Household members (base: two or less)	
three or more	-0.164#
Work-related variables	
Work schedule flexibility (base: Inflexible)	
Somewhat flexible	0.228*
Tech-savviness related variables	
Smart home usage frequency (base: Never)	
Frequent (everyday)	0.787***
Infrequent (once a week to a year)	0.624***
GPS navigation usage frequency (base: Never)	
Frequent (everyday)	-0.306*
Infrequent (once a week to a year)	-0.489***
CAV related variables	
WTP for installing autonomous technology in existing conventional car (base: < $2,500)	
more than $7,500	0.311**
WTP for annual maintenance of AV technology (base: $0)	
more than $300	-0.183#
Familiar with AV technology (base: No)	
Yes	-0.765***
Interested to privately own a CAV (base: No)	
Yes	-0.86***
Interested to use ride hailing CAV with a backup driver (base: No)	
Yes	-0.859***
Interested to use ride hailing CAV with no backup driver (base: No)	
Yes	0.318*
Involved in vehicle crashes in the past (base: No)	
Yes	-0.48***
Vehicle driving related variables	
Driving mileage (annual) (base: less than 5,000 miles)	
more than 5,000 miles	-0.456***
Physical disability undermining driving (base: No)	
Yes	0.742***
Survey channel related variables	
Survey distribution channel (base: Panel from market research company)	
Amazon Mechanical Turk (MTurk)	-0.866***
Social media and educational institutes	-0.826***

Goodness of fit measure: same as Table 4.**Significance levels:** -- not significant, #0.10, *0.05, **0.01, ***0.001

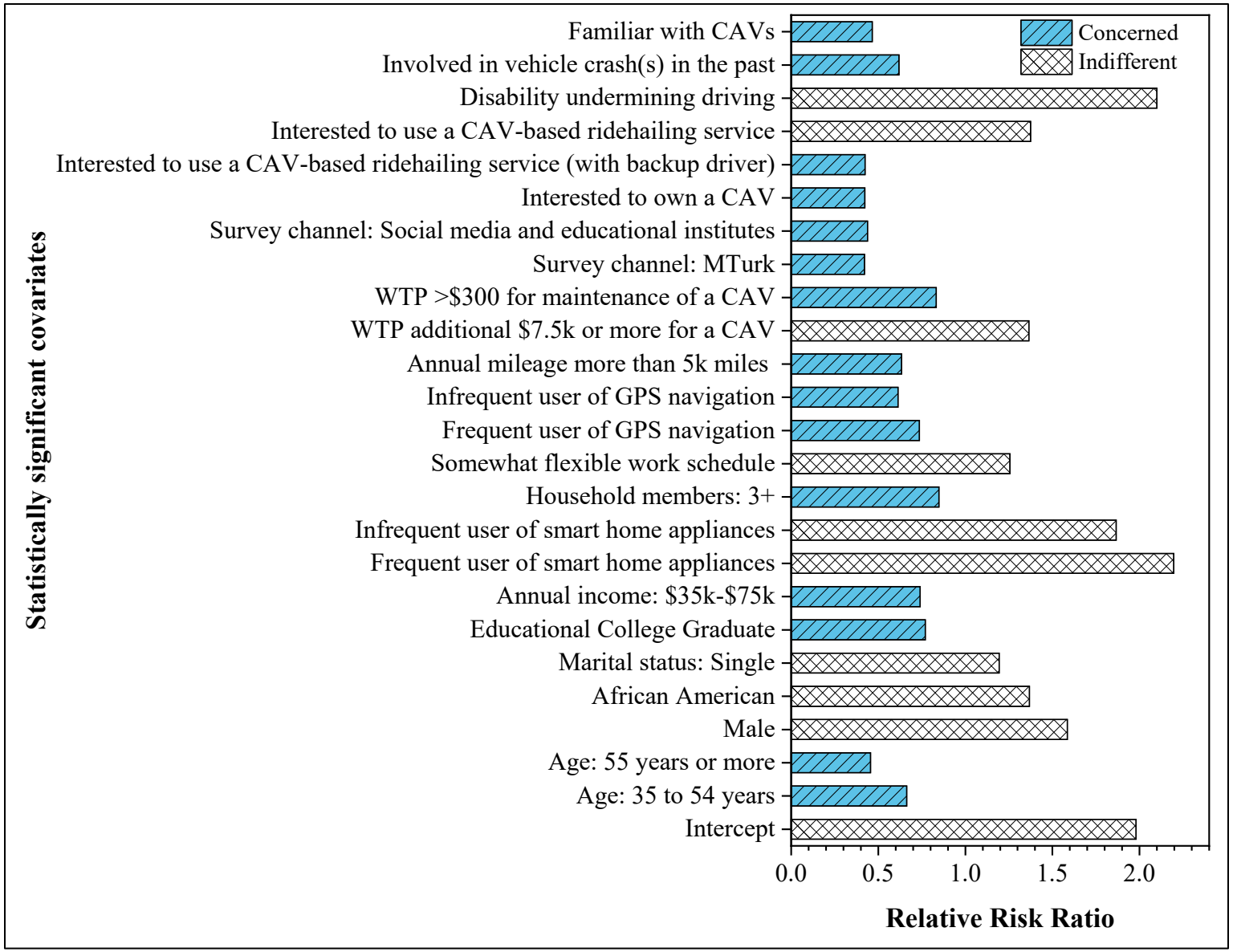


Fig. 5. Class membership of the covariates based on relative risk ratio (RRR)

For brevity, we discuss results where coefficients were significant for both latent classes. Overall, signs are the same among all such estimates in both classes. However, the latent class Indifferent experiences a stronger effect on a particular information source's (un)reliability. Compared to females, males are less likely to rely on advertisements and car dealers because males spend more time on personal research and are less influenced by advertisements (Facebook, 2018). It can also be attributed to the different lifestyles of make and females and the propensity of females to have more personal interactions (van Rijnsoever et al., 2012). **Highly qualified** individuals are more likely to rely on the input received from their peers. It can be attributed to their propensity to obtain CAV-related information from their peers, as well-educated individuals are more open to technological innovations (Talukder et al., 2019). Such individuals are more likely to depend on limited information sources consistent with existing literature (Kim and Ratchford, 2012b; Klein and Ford, 2003; Ratchford et al., 2007, 2003)

Interestingly, individuals having at least one **peer enthusiastic about the cars** are more likely to rely on all four information channels, and this effect is stronger for three or more peers. As expected, the effect is higher for the Friend/Colleague information source. Such individuals will prefer these car enthusiastic peers over others. Consequently, an **increased number of peers** was positively related to car dealers' reliability, which highlights the importance of car enthusiastic peers among these peers. Individuals **willing to pay $15,000 to $30,000 for a conventional vehicle** are more likely to trust car dealers for the information. Such residents might buy a new/used car from the nearest car dealer, not third parties. Compared to respondents hired from the market research company panel, MTurk respondents are more likely to rely on personal research and their peers, which can be attributed to the high proportion of young, highly educated, and low-income earners in MTurk (Hitlin, 2016).

Residents **interested in adopting a privately owned CAV** are more likely to rely on personal research and their peers to receive CAV-related information. Interestingly, Concerned individuals are more likely to prefer peers over their personal research as word of mouth is related positively to decreasing the associated risk in product adoption (Hussain et al., 2018, 2017; Manning et al., 1995; Tan, 1999). Individuals interested in **CAVs capability to make automated service trips** prefer both personal research and car dealers. However, for the Concerned and Indifferent, the effect was stronger for personal research and car dealers, respectively. It can be attributed to concerned individuals' propensity to browse online blogs to confirm the feasibility of such features in CAVs, even if they rely on car dealers for the information (Jeong et al., 2017). As expected, individuals **interested in sharing/carpooling a CAV** are more likely to rely on their peers for the CAV information. Individuals interested in riding in a CAV-based ride-hailing service without a backup driver are more likely to rely on media advertisements which can be attributed to the success of the recent pilot program of Google Waymo on the public road (Waymo, 2020).

5.2.1 Omnichannel behavior

Based on the estimated coefficients, we also identify the characteristics of the individuals who rely on more than one information source. Synonymous with existing literature on conventional automobile purchases (Dahiya and Gayatri, 2018; Jang et al., 2017), we found that consumers rely on more than one source. Individuals willing to pay $0 to $300 for annual maintenance of CAVs, interested in owning a CAV, interested in a CAV's ability to make automated service station trips, and having peers who are knowledgeable about cars are more likely to rely on all four sources. Such finding highlights the consumers' increased interest in CAVs linked with searching for all channels despite having car enthusiastic peers in their social network.

Frequent TV watchers, willing to pay higher for CAVs and interested in CAV-based ride-

hailing services with no human driver are more likely to rely on advertisements and personal research. Such finding highlights the importance of the success of pilot projects of CAV-based ride-hailing services on CAV acceptance. Consumers willing to pay $15,000 to $30,000 for a new conventional car are more likely to depend on car dealerships, advertisements, and personal research, which is as expected as such individuals will make an effort to get the best value for their money after consulting car dealers, looking for offers in advertisements in addition to their personal research over the internet.

Highly qualified, familiar with CAVs, and interested in carpooling a CAV are more likely to depend on their personal research and peers. African Americans and an increased number of peers are more likely to rely on car dealerships and advertisements to obtain CAV-related information. Consumers willing to pay higher for a conventional car rely on car dealerships and personal research. Such consumers are more likely to visit the nearest car dealership to buy the car they found through their research as the price is no bar for them. Frequent radio listeners rely on peers, advertisements, and car dealerships.

In addition to the estimated coefficients of four information sources, we also explore the omnichannel behavior of these sources based on their error correlation (Table 6). All sources **positively correlate with one another,** implying that individuals rely on more than one source to collect information (Dahiya and Gayatri, 2018; Jang et al., 2017). Unsurprisingly, the maximum correlation was between advertisements and car dealers. Both sources offer similar information (attractive features and price). Similarly, personal research and peer-related sources also include high correlation as individuals look to verify their information with their peers and hence rely on both sources.

5.3 Marginal effects

To discuss policy implications, we focus on the marginal effects for latent classes, Concerned and Indifferent, portrayed in Fig. 6 and Fig. 7, respectively. For brevity, we did not present the marginal effects for the "Neutral" category and only for the "Reliable" and "Unreliable" categories. The y-axis represents the statistically significant covariates with their description. The X-axis represents the marginal effects (already multiplied by 100). A positive (negative) value can be interpreted as: ceteris paribus, one unit change in a particular covariate will increase (decrease) the likelihood of a particular level of the dependent variable by a specified percentage (magnitude of the effect). For instance, for the dependent variable Friend/Colleague for Concerned latent class, an interest in owning a car will increase the individuals' likelihood of relying on their friends/colleagues for the information increases by about 11%. We only discuss the results for the dependent variable category "Reliable" presented in Fig. 6 and Fig. 7 as it will be more insightful for framing policies. Based on estimated coefficients bolstered by their marginal effects, this research offers numerous policies for individuals concerned about the negative impacts of CAVs, discussed in the following subsections for each information source.

Table 6. Multivariate ordered logit: results (N: Class 1 = 3,417, Class 2 = 1,031)

Variable	Coefficient^{Significance}							
	Friend/Colleague		Advertisements		Car Dealers		Personal Research	
	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2
Intercept	0.369	0.585*	-0.286	0.457$^{\#}$	-0.207	0.149	0.17	0.444$^{\#}$
Socioeconomic variables								
Age (base: 18 to 35 years)								
35 to 54 years	--	--	--	0.431**	0.214*	--	--	--
more than 54 years	--	--	--	--	0.249*	--	--	--
Gender (base: Female)								
Male	-0.156$^{\#}$	--	-0.37***	-0.275*	-0.509***	-0.356**	--	--
Ethnicity (base: others)								
African American	--	--	0.514***	--	0.681***	--	--	--
Marital status (base: Separated, divorced, widowed)								
Single	--	--	-0.202*	--	--	--	--	--
Educational attainment (base: High school or below)								
Some College	--	--	-0.321**	--	-0.294**	--	0.417***	--
College Graduate	0.288*	--	-0.214*	--	-0.254*	--	0.51***	--
Master's, doctoral or professional degree	0.442**	0.592***	-0.218$^{\#}$	--	-0.354**	--	0.759***	--
Household-related variables								
Household locality (base: Rural)								
Urban	-0.231*	--	--	--	--	--	-0.29**	--
Vehicle ownership related variables								
Household cars (base: one and zero)								
three or more	--	--	--	-0.381*	--	-0.378*	--	--
New cars in the household (last ten years) (base: one and zero)								
three or more	--	--	--	--	0.182*	--	--	--
Used cars in the household (last ten years) (base: zero)								
one	--	0.31*	--	0.252$^{\#}$	-0.21*	--	--	--
two or more	--	--	-0.264**	--	-0.407***	--	--	--
Conventional car purchase frequency (base: Once a year to every 2-3 years)								
Once every 15 to 20 years	--	--	-0.246*	--	-0.257*	--	--	--
Willingness to pay for a conventional car (base: less than $15,000)								
$15,000-$30,000	--	--	0.226**	--	0.295***	0.385*	0.217*	--
more than $30,000	--	--	--	--	0.385***	--	0.331*	--
Driving mileage (annual) (base: less than 5,000 miles)								
more than 5,000 miles	0.154$^{\#}$	--	--	--	--	--	0.218*	--
Physical disability undermining driving (base: No)								
Yes	-0.292*	--	--	--	0.233$^{\#}$	--	--	--

Table 6 continued

Variable	Coefficient[Significance]							
	Friend/Colleague		Advertisements		Car Dealers		Personal Research	
	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2
Work and daily commute-related variables								
Remote work frequency (base: Never)								
Frequent (everyday)	--	--	--	--	--	--	-0.311*	--
Infrequent (once a week to a year)	--	--	--	-0.439*	--	--	--	--
Work schedule flexibility (base: Inflexible)								
Somewhat flexible	--	--	--	--	--	0.335#	--	--
Private car usage frequency (base: Never)								
Frequent (everyday)	-0.361**	--	-0.31*	--	-0.255*	--	--	--
Infrequent (once a week to a year)	-0.877***	--	--	--	--	--	--	--
Public transport usage frequency (base: Never)								
Frequent (everyday)	--	-0.616**	--	-0.634**	--	-0.96***	--	-0.399#
Ridesharing service usage frequency (base: Never)								
Frequent (everyday)	--	-0.806**	--	--		-0.437#	--	-0.793***
Ride hailing service usage frequency (base: Never)								
Frequent (everyday)	--	-0.666**	--	-0.629*	--	--	--	-0.72**
Infrequent (once a week to a year)	--	--	--	0.44#	--	--	--	--
Tech-savviness related variables								
Radio listening frequency (base: Never)								
Frequent (everyday)	0.428*	--	0.386*	--	0.409*	--	--	--
Infrequent (once a week to a year)	0.424*	--	--	--	--	--	--	--
TV watching frequency (base: Never)								
Frequent (everyday)	--	--	0.595*	--	--	--	0.694*	--
Infrequent (once a week to a year)	--	--	--	--	--	--	0.635#	--
Smart home usage frequency (base: Never)								
Frequent (everyday)	-0.217*	--	0.181*	--	--	--	--	--
Infrequent (once a week to a year)	--	--	--	0.29#	--	--	--	--
CAV related variables								
WTP for installing autonomous technology in existing conventional car (base: < $2,500)								
$2,500 to $7,500	0.243*	--	0.226*	--	--	--	0.208#	--
more than $7,500	0.363**	--	0.349***	--	--	--	0.245#	--
WTP for annual maintenance of AV technology (base: $0)								
$0 to $300	0.259*	--	0.297**	--	0.284**	--	0.374**	0.259#
more than $300	--	--	0.356***	--	0.424***	--	0.32*	--
Familiar with AV technology (base: No)								
Yes	0.288**	--	--	--	-0.17#	--	0.394***	--
WTP towards automated service station trips in CAVs (base: No)								
Yes	0.268*	--	--	0.618***	0.18*	0.487**	0.269*	0.373*
Interested to privately own a CAV (base: No)								
Yes	0.571***	0.736***	0.362***	--	0.168#	--	0.368**	0.744***

Table 6 continued

Variable	Coefficient^{Significance}							
	Friend/Colleague		**Advertisements**		**Car Dealers**		**Personal Research**	
	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2
Interested to use Autonomous Public Transport service (base: No)								
Yes	--	-0.307$^{\#}$	0.162$^{\#}$	--	0.162$^{\#}$	--	--	--
Interested to use ride hailing CAV with a backup driver (base: No)								
Yes	--	--	--	--	0.211*	--	--	--
Interested to use ride hailing CAV with no backup driver (base: No)								
Yes	--	--	0.493***	0.533**	--	--	--	0.429*
Interested to carpool/share a CAV (base: No)								
Yes	0.312*	0.591***	--	--	--	--	--	0.397*
Involved in vehicle crashes in the past (base: No)								
Yes	--	--	--	-0.477***	--	-0.296*	--	--
Peer network related variables								
Peers knowledgeable about cars (base: zero)								
1 to 2 peers	0.525***	0.727***	0.207*	0.346$^{\#}$	0.414***	0.461**	0.284*	0.366*
more than 2 peers	0.612***	0.807***	0.321***	0.638***	0.335***	0.657***	0.365***	0.676***
Peers' communication frequency (base: Infrequent, i.e., once per month to every few months)								
Frequent (2-3 times a week to daily)	0.391***	--	0.184$^{\#}$	--	--	--	0.293**	0.452***
Sometimes (every couple of weeks to a month)	0.262*	--	--	--	--	--	0.242$^{\#}$	--
Number of established peers	--	-0.007*	0.004**	--	0.005***	0.01***	-0.007***	--
Survey channel related variables								
Survey distribution channel (base: Panel from market research company)								
Amazon Mechanical Turk (MTurk)	0.626***	0.929***	-0.197*	--	-0.451***	--	0.861***	0.794**
Social media and educational institutes	0.65***	--	-0.323**	--	-0.486***	--	0.318$^{\#}$	--
Thresholds								
Threshold 1 (Neutral/Unreliable)	0.0***	0.0***	0.0***	0.0***	0.0***	0.0***	0.0***	0.0***
Threshold 2 (Reliable/Neutral)	1.618***	2.414***	1.493***	2.339***	1.232***	2.172***	1.475***	2.097***
Error correlations								
Friend/Colleague	--	--						
Advertisements	0.291***	0.443***	--	--				
Car Dealers	0.173***	0.364***	0.634 ***	0.612***	--	--		
Personal Research	0.385***	0.380***	0.322***	0.420***	0.293***	362***	--	--

Goodness of fit measures: Class 1 Concerned - log-likelihood= -33,466; AIC = 68,123; BIC = 71,774
Class 2 Indifferent - log-likelihood= -11,410; AIC = 23,716; BIC = 25,927
Significance levels: -- not significant, #0.10, *0.05, **0.01, ***0.001

5.3.1 Latent Class: Concerned

As per Fig. 6a, the likelihood of reliability on peers is increased by almost 12%, 8%, 9%, and 8% for individuals with three or more peers knowledgeable about cars, frequently communicating with their peers, frequent radio listeners, and with at least a post-graduate degree, respectively. Individuals' likelihood to prefer advertisements (Fig. 6b) increases (marginal effect in brackets) if they are frequent TV watchers (13%), African Americans (12%), interested in CAV-based ride-hailing services (12%), frequent radio listeners (9%), willing to pay higher for CAVs (8%) and physically challenged (5%). Similarly, inferred from the marginal effects of the car dealer information source in Fig. 6c, it is evident that individuals willing to pay higher for conventional cars, bought three or more cars in the last ten years, and individuals aged over 35 years increase the dependence likelihood by 10%, 5%, and at least 5%, respectively. Lastly, as per Fig. 6d, for personal research, individuals who are familiar with CAVs, interested in owning a CAV, college graduates or higher, and with higher annual vehicle mileage increase the dependency likelihood by 7%, 6%, at least 6%, and 3%, respectively.

5.3.2 Latent Class: Indifferent

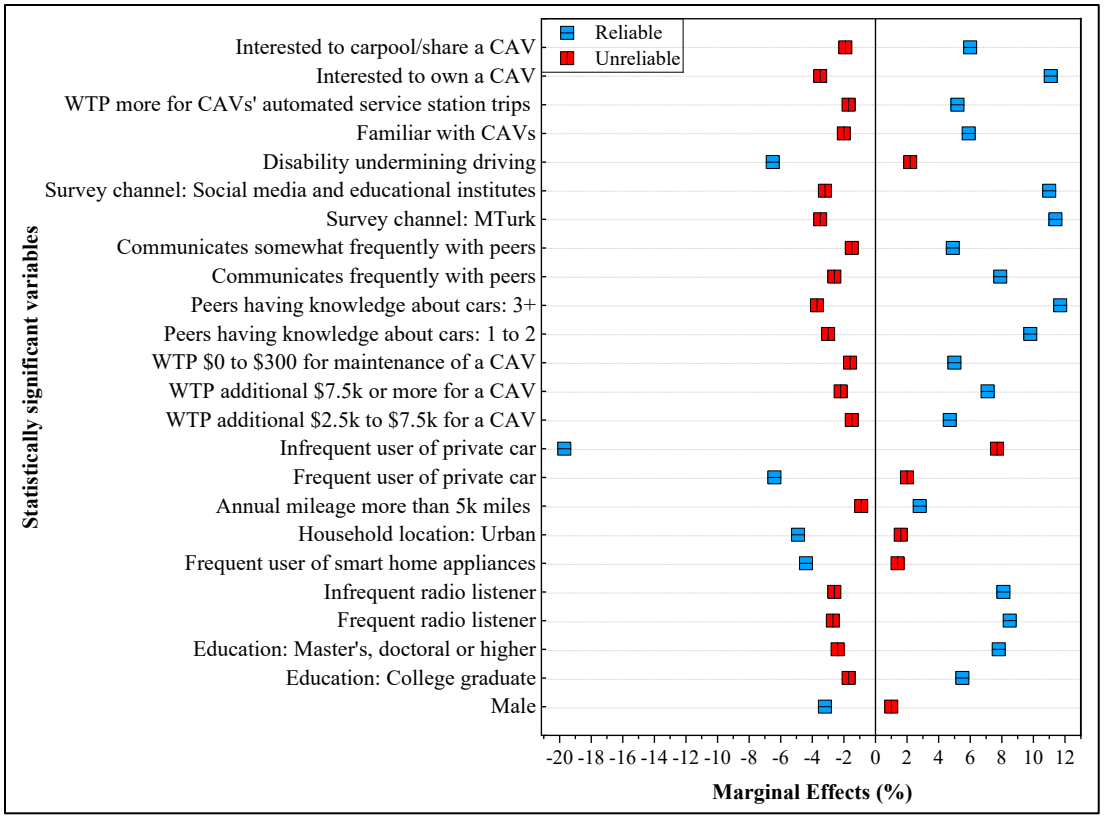
As per Fig. 7a, Indifferent individuals who are highly qualified, interested in carpooling a CAV, and bought one used car in the last ten years are likely to increase the likelihood of relying on their peers by 12%, 12%, and 6%, respectively. Individuals' likelihood to depend on advertisements (Fig. 7b) increases by 7%, 8%, and 9% if they are interested in CAV-based ride-hailing services, infrequent users of ride-hailing services, and aged between 35 to 54 years, respectively. Similarly, as per Fig. 7c, individuals' likelihood to rely on car dealerships will increase by 7%, 6%, and 6% if they are willing to pay between $15,000 to $30,000 for a conventional car, following a somewhat flexible work schedule, and involved in vehicle crashes in the past. Finally, individuals' likelihood to obtain information through personal search, presented in Fig. 7d, increases by 11%, 10%, and 6% if they are frequently communicating with their peers, interested in CAV-based public transport, and willing to pay $0 to $300 for annual maintenance of CAVs.

6. Policy implications

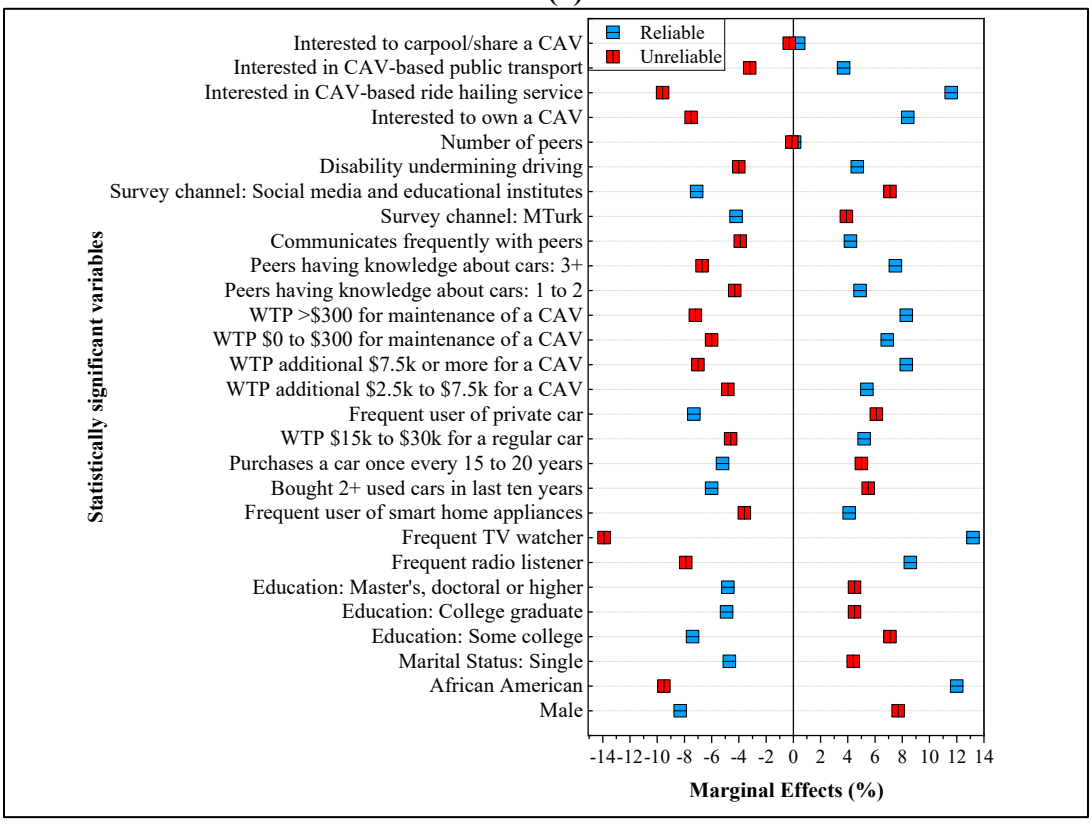
This research offers novel implications for individuals concerned about the negative aspects of CAVs and their dependence on multiple sources of information before finalizing the decision to purchase a CAV. Based on estimated coefficients bolstered by their marginal effects, this research offers numerous policies for individuals concerned about the negative impacts of CAVs. All such policies will provide a novel framework for involved stakeholders to identify appropriate information strategies to disseminate CAV-related information in boosting their wider acceptance.

6.1 Managerial implications for resolving negative impacts of CAVs

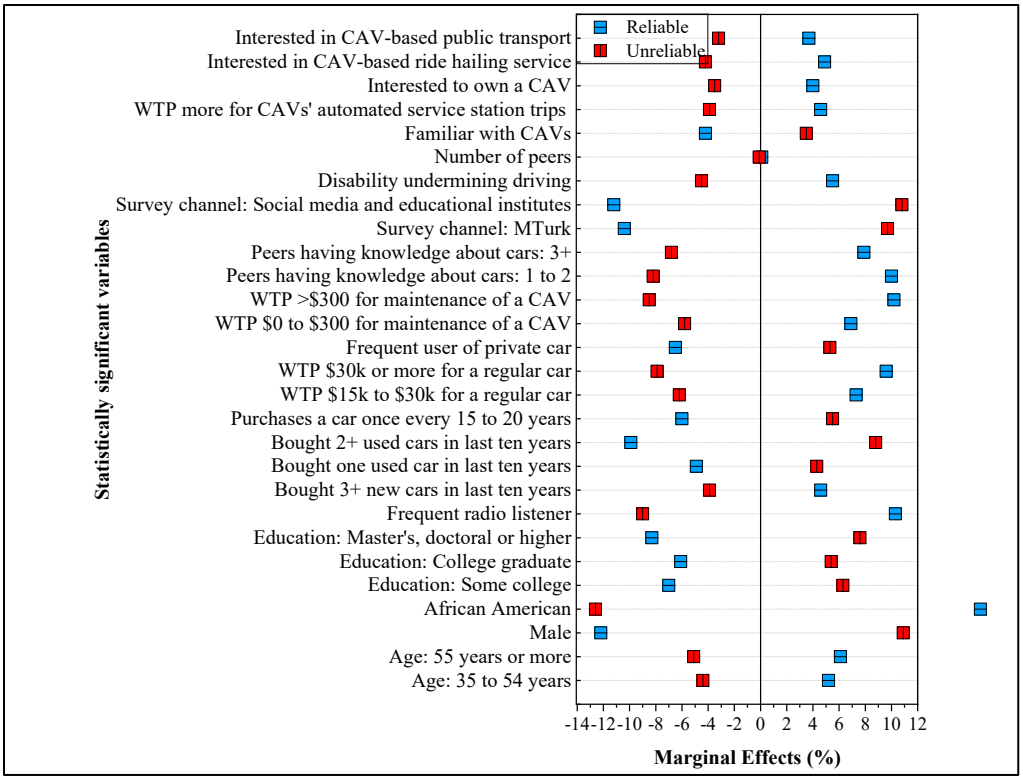
Highly qualified, frequent radio listeners, willing to pay more for CAVs, including an increased number of car enthusiasts in their peer network, communicating regularly with peers, familiar with CAVs, and interested in owning a CAV are more likely to be influenced by CAV information received from their peers. *Such findings provide automakers and car dealers to increase their outreach on social media while emphasizing the benefits and solutions for the identified barriers of CAVs.* Such agencies can also offer a referral-based incentive to early adopters of CAVs to encourage them to share their experiences with their peers in the social network (both work and non-work).


Reliable
Unreliable
Interested to carpool/share a CAV
Interested to own a CAV
WTP more for CAVs' automated service station trips
Familiar with CAVs
Disability undermining driving
Survey channel: Social media and educational institutes
Survey channel: MTurk
Communicates somewhat frequently with peers
Communicates frequently with peers
Peers having knowledge about cars: 3+
Peers having knowledge about cars: 1 to 2
WTP $0 to $300 for maintenance of a CAV
WTP additional $7.5k or more for a CAV
WTP additional $2.5k to $7.5k for a CAV
Infrequent user of private car
Frequent user of private car
Annual mileage more than 5k miles
Household location: Urban
Frequent user of smart home appliances
Infrequent radio listener
Frequent radio listener
Education: Master's, doctoral or higher
Education: College graduate
Male
Statistically significant variables
-20 -18 -16 -14 -12 -10 -8 -6 -4 -2 0 2 4 6 8 10 12
Marginal Effects (%)

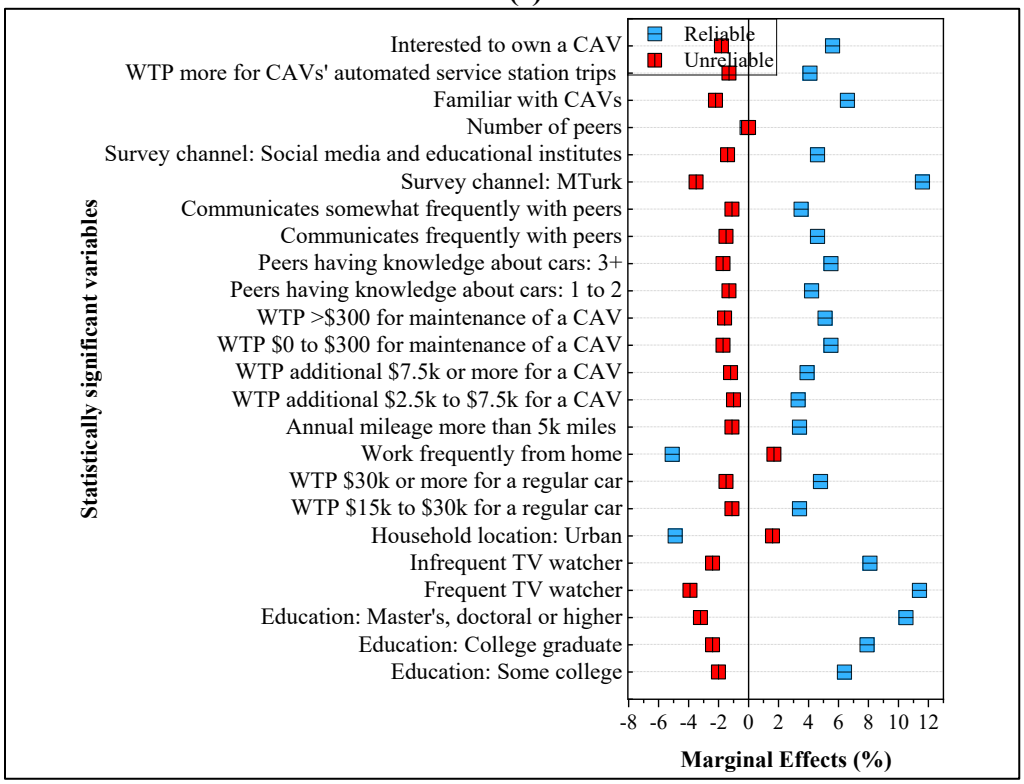
(a)


Reliable
Unreliable
Interested to carpool/share a CAV
Interested in CAV-based public transport
Interested in CAV-based ride hailing service
Interested to own a CAV
Number of peers
Disability undermining driving
Survey channel: Social media and educational institutes
Survey channel: MTurk
Communicates frequently with peers
Peers having knowledge about cars: 3+
Peers having knowledge about cars: 1 to 2
WTP >$300 for maintenance of a CAV
WTP $0 to $300 for maintenance of a CAV
WTP additional $7.5k or more for a CAV
WTP additional $2.5k to $7.5k for a CAV
Frequent user of private car
WTP $15k to $30k for a regular car
Purchases a car once every 15 to 20 years
Bought 2+ used cars in last ten years
Frequent user of smart home appliances
Frequent TV watcher
Frequent radio listener
Education: Master's, doctoral or higher
Education: College graduate
Education: Some college
Marital Status: Single
African American
Male
Statistically significant variables
-14 -12 -10 -8 -6 -4 -2 0 2 4 6 8 10 12 14
Marginal Effects (%)

(b)

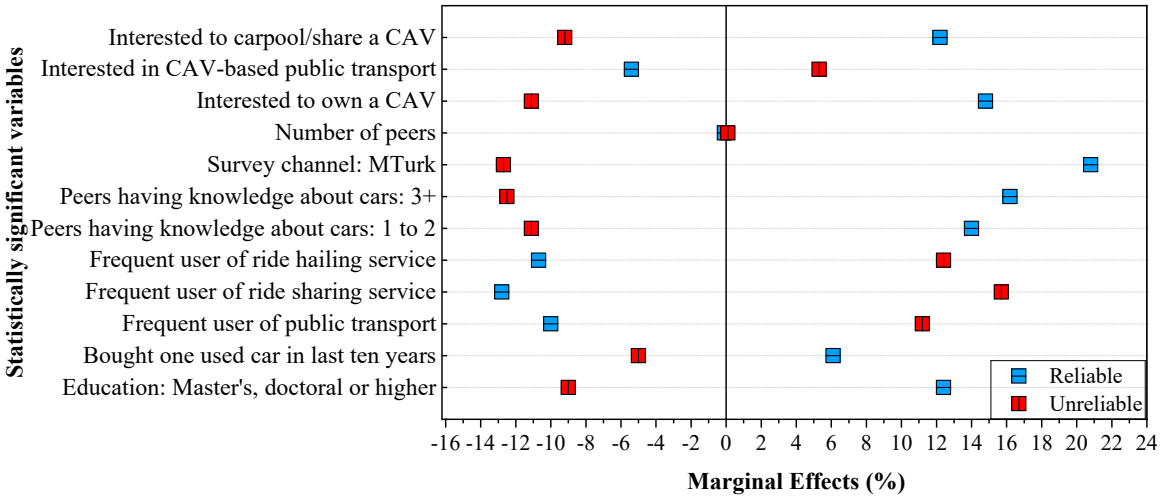


(c)

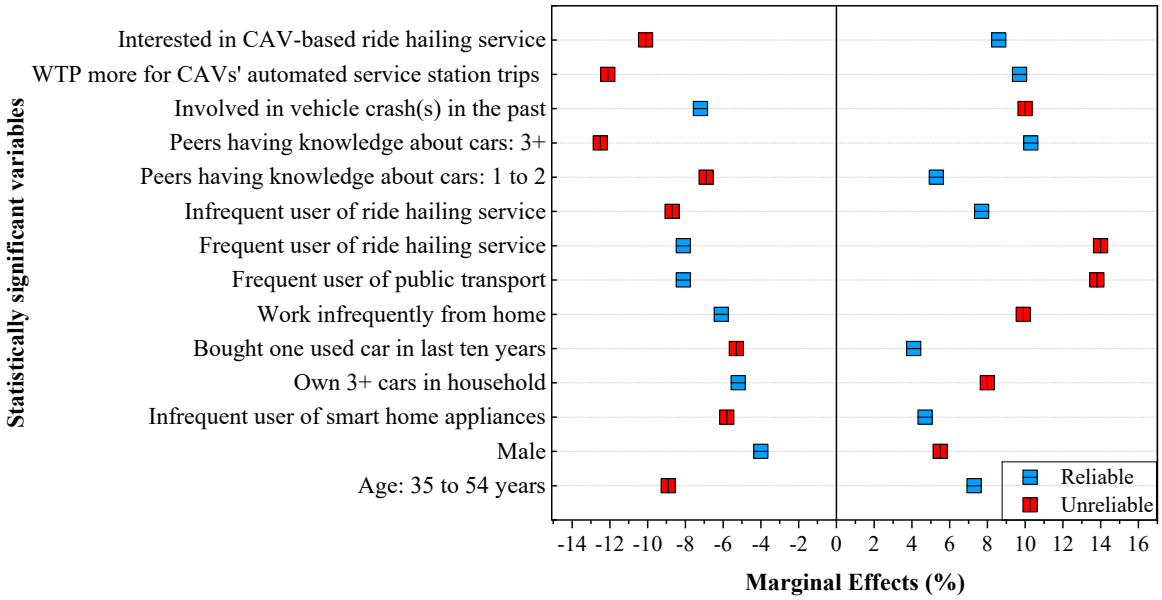


(d)

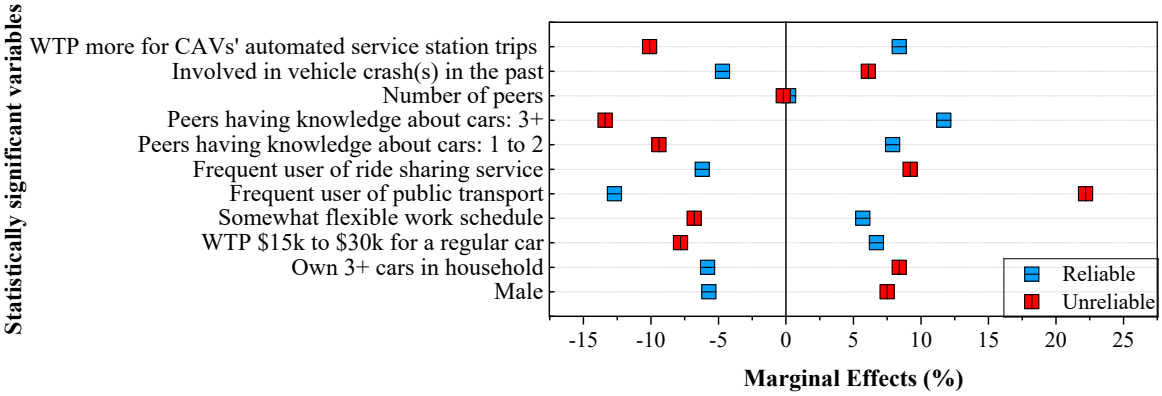
Fig. 6. Marginal effects for latent class Concerned (a) Friend/Colleague (b) Advertisements (c) Car Dealers (d) Personal Research


Statistically significant variables
Interested to carpool/share a CAV
Interested in CAV-based public transport
Interested to own a CAV
Number of peers
Survey channel: MTurk
Peers having knowledge about cars: 3+
Peers having knowledge about cars: 1 to 2
Frequent user of ride hailing service
Frequent user of ride sharing service
Frequent user of public transport
Bought one used car in last ten years
Education: Master's, doctoral or higher
Reliable
Unreliable
-16 -14 -12 -10 -8 -6 -4 -2 0 2 4 6 8 10 12 14 16 18 20 22 24
Marginal Effects (%)

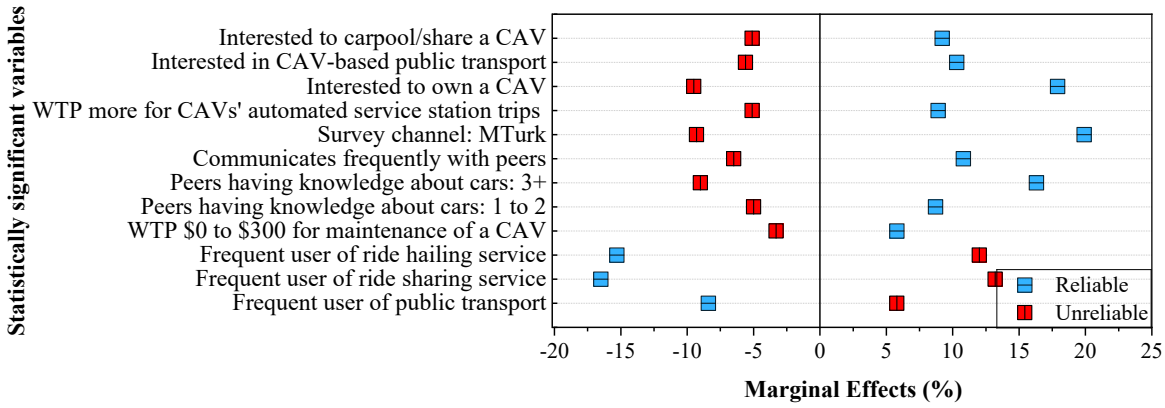
(a)


Statistically significant variables
Interested in CAV-based ride hailing service
WTP more for CAVs' automated service station trips
Involved in vehicle crash(s) in the past
Peers having knowledge about cars: 3+
Peers having knowledge about cars: 1 to 2
Infrequent user of ride hailing service
Frequent user of ride hailing service
Frequent user of public transport
Work infrequently from home
Bought one used car in last ten years
Own 3+ cars in household
Infrequent user of smart home appliances
Male
Age: 35 to 54 years
Reliable
Unreliable
-14 -12 -10 -8 -6 -4 -2 0 2 4 6 8 10 12 14 16
Marginal Effects (%)

(b)


Statistically significant variables
WTP more for CAVs' automated service station trips
Involved in vehicle crash(s) in the past
Number of peers
Peers having knowledge about cars: 3+
Peers having knowledge about cars: 1 to 2
Frequent user of ride sharing service
Frequent user of public transport
Somewhat flexible work schedule
WTP $15k to $30k for a regular car
Own 3+ cars in household
Male
Reliable
Unreliable
-15 -10 -5 0 5 10 15 20 25
Marginal Effects (%)

(c)



(d)

Fig. 7. Marginal effects for latent class Indifferent (a) Friend/Colleague (b) Advertisements (c) Car Dealers (d) Personal Research

Physically challenged, African Americans, frequent TV watchers, tech-savvy individuals willing to pay higher for CAVs and interested in using a ride-hailing CAV are more likely to rely on advertisements. Such finding is beneficial to invest in the targeted advertising, focusing on ride-hailing CAVs' success on print and media advertisements. Automakers can use voice-enabled smart assistants (tech-savvy behavior) to integrate CAV-related announcements. The car dealers and third-party companies can target such individuals (after accepting the appropriate consent). Older individuals, females, African Americans, buying three or more cars in the last ten years, paying higher for conventional cars, interested in automated service station trip making capability of CAVs, and interested in using a ride-hailing CAV with a backup driver are more likely to rely on car dealers for the information. Such findings allow car dealers to focus more on the operational aspect of CAVs and include such *information in pamphlets or flyers at the dealerships.* Well-qualified, frequent TV watchers, driving more than 5,000 miles annually, individuals with limited social contacts and interested in owning or sharing a CAV are more likely to rely on their research. Such findings highlight the *opportunity for third-party sources like Edmunds and Carfax to include CAV-related information threads on their websites, blogs, and YouTube channels.* Car manufacturers are also encouraged to provide CAV-related information on their websites and third-party blogs based on these findings. Such policies will help resolve residents' concerns about CAV barriers and boost the acceptance rate when implemented.

6.2 Omnichannel CAV buyers

We also extract policy implications from individuals who depend on more than one information source. Overall, consumers' increased interest in CAVs and their anticipated features is linked to consulting all four sources. In addition, consumers who buy mid-segment regular cars are more likely to rely on all sources except peers. These findings highlight the *need for stakeholders involved in CAV sales to focus on multi-channel information dissemination*. Based on error correlations, consumers are more likely to rely on car dealerships and advertisements followed by peers-personal research. Hence stakeholders can prioritize information dissemination through these channels. Based on the omnichannel results, we also offer key insights for stakeholders interested in using the most effective communication channel for the CAV information dissemination. In that context, stakeholders should prioritize media advertisements over other sources as the majority of the covariates relate to the advertisement as the source of information.

In addition, advertisements can emulate car dealerships to disseminate the information. Furthermore, internet advertisements should be preferred over other forms as they might cover individuals trusting their personal research (through blogs and videos).

7. Conclusions

Connected and Autonomous Vehicles (CAVs) have the potential of transforming the way humans complete their travel needs. Recent research in this direction vindicates the imminent penetration of CAVs through either replacing or complementing the existing conventional vehicles. However, their widespread acceptance is still questionable as CAVs include numerous barriers to their adoption, i.e., less user acceptance, safety concerns, cybersecurity issues, and legality problems. Recent research indicates a greater influence of vehicle-related information obtained from four primary sources (interpersonal communication, car dealerships, mass media advertisements, and personal research) on consumers' decisions to finalize vehicle purchases. The research also indicates the anticipated impact of such sources, especially interpersonal communication and mass media advertisements, on the changing behavior of individuals towards autonomous technology. Hence building on the literature on negative aspects of CAVs and the impact of information sources on conventional vehicle purchases, we contribute to the existing literature by quantifying the potential consumers' dependence on four information sources before purchasing a CAV. Utilizing a two-step econometric modeling framework, latent class analysis followed by multivariate ordered logit, and a statewide survey, we also compare this quantification among individuals skeptical and indifferent about CAVs.

The results indicated that the elderly, three or more person households, and residents interested in owning a CAV are more likely to be concerned about negative aspects of CAVs. Results also showed that Concerned and Indifferent individuals have similar behavior toward a relevant information source, the latter showing a more robust association. Consistent with conventional vehicle literature, individuals relied on more than one source. Individuals interested in owning a CAV are more likely to rely on their peer network before finalizing their decision. We also offer several policies for disseminating CAV-related information to boost consumer acceptance. Automakers and car dealers are encouraged to increase their outreach on social media to disseminate information related to identified barriers to CAVs and offer a referral-based incentive for CAV buyers to share their experiences among their peers. Stakeholders are encouraged to invest in targeted advertisements (for physically challenged, African Americans, frequent TV watchers, and tech-savvy individuals), focusing on the success of ride-hailing CAVs. Car dealerships can focus more on the operational aspect of CAVs through pamphlets. Findings from personal research-related information sources highlight the opportunity for third-party sources like Edmunds and Carfax to include CAV-related information threads on their websites, blogs, and YouTube channels. Such policies will help resolve residents' concerns about CAV barriers and boost the acceptance rate when implemented. Such findings will be helpful for policymakers, automakers, car dealerships, and third-party agencies involved in CAVs to advertise, promote and sell CAVs through appropriate information channels.

8. Limitations

Our research also includes some limitations in terms of insufficient evidence for the individuals unconcerned or excited about CAVs. Furthermore, the survey overrepresented residents with high-

income levels, high educational attainment, and internet access in their households. Hence the findings should be considered with caution. This research also offers some future avenues of research. We explored the qualitative importance (behavioral) of principal information sources on CAV acceptance but not the qualitative (market penetration levels). The important findings related to the presence of automobile enthusiast peers in individuals' social networks and frequent communication with their peers call for a need to explore the studies exploring the effect of such peers on the market adoption rate of CAVs. Future studies can also explore the role of positive and negative information received from all these sources on CAV acceptance, especially for individuals unconcerned about the barriers associated with CAVs. Future efforts can also use robust estimators as part of robustness analysis.

As per the existing literature, choice behavior suffers from status quo bias or reference effect (Masatlioglu and Ok, 2005; Masatlioglu and Uler, 2013; Wang et al., 2021). Status quo bias refers to the increased likelihood of a respondent choosing an alternative when the alternative is the default option (Masatlioglu and Ok, 2005). On the other hand, the reference effect captures the impact of certain reference points (default alternatives) on the decision maker's choice behavior even if the decision-maker does not choose the reference point (Masatlioglu and Uler, 2013). Due to the absence of a choice experiment capturing the impact of different attributes of information sources and capturing the past behavior of pre-purchase information search, this study did not consider the impact of consumer's self-reference effect or status quo bias on considering information sources during the CAV pre-purchase phase (Masatlioglu and Uler, 2013; Wang et al., 2021). Future research can account for such reference or status-quo effect while utilizing a discrete choice experiment focusing on multiple information sources in the survey questionnaire (Kim et al., 2020; Wang et al., 2021). Future avenues of research can also explore establishing a reference point for selecting information source alternatives (Wang et al., 2021) based on the past pre-purchase behavior of individuals toward conventional vehicles.

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Author Contributions

The authors confirm contributions as follows: study conception and design: Ishant Sharma and Sabyasachee Mishra; data collection: Ishant Sharma and Sabyasachee Mishra; analysis and results: Ishant Sharma; draft manuscript preparation: Ishant Sharma and Sabyasachee Mishra. All authors reviewed the results and approved the final version of the manuscript.

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